

Feedback — Hard Negative Mining for KG Link Prediction

1. Overall assessment

This is one of the clearest Milestone 2 submissions. The project has a precise research question, a controlled ablation, strong baselines, learning curves, filtered ranking metrics, slice evaluation, qualitative examples, limitations, and concrete member contributions. The evidence convincingly supports the claim that hard-negative selection improves RotatE link prediction on FB15k-237 under the tested conditions. Before final delivery, the main work is polish: clarify what is fully controlled and avoid overgeneralizing beyond RotatE and the chosen pool size.

2. What is already good

- The research question is sharp and testable.
- The negative-sampling pipeline is modular and well described.
- The ablation controls all major hyperparameters across strategies.
- Results include MRR, Hits@k, training curves, slice analysis, and qualitative examples.
- Limitations are explicit, especially the custom-pipeline versus PyKEEN comparison.
- Member contributions are concrete.

3. Main issues to fix before the final delivery

1. **Issue:** The comparison with PyKEEN baselines is not fully controlled.
Why it matters: The strongest causal claim should come from the custom strategy ablation, not the PyKEEN comparison.
Concrete action: In the final report, phrase the PyKEEN comparison as context and make Table 3 the central controlled result.
2. **Issue:** Pool size $n=64$ is fixed and not ablated.
Why it matters: The hard-negative effect may depend on candidate-pool size.
Concrete action: State this as a limitation; if time permits, run only one extra pool-size check for the best mixed strategy.
3. **Issue:** All custom experiments use RotatE.
Why it matters: The conclusion may not transfer to TransE or other KGE models.
Concrete action: Keep claims specific to RotatE unless you run one lightweight TransE hard/random comparison.
4. **Issue:** The poster may become table-heavy.
Why it matters: The main idea is easy to visualize.

Concrete action: Add a simple diagram of pool generation, candidate scoring, and selection strategies.

4. Milestone 2 requirements checklist

- Training methods: **OK**.
- Experiments: **OK**.
- Results: **OK**.
- Learning curves: **OK**.
- Contribution of each group member: **OK**.

5. Suggestions from M1 that still need attention

The M1 feedback asked the group to turn the project into a clear scientific contribution rather than only reporting RotatE versus TransE. This was addressed very well through the controlled hard-negative ablation, slice evaluation, and qualitative analysis. No major M1 item remains unresolved.

6. Recommended final experiments or analyses

Must do: - Keep the controlled custom ablation as the central result and label the PyKEEN comparison carefully. - Add a clear limitations paragraph on RotatE-only and fixed pool size.

Nice to have: - Run a single extra ablation for pool size or TransE if it can be done quickly. - Add a small runtime/cost table for random versus hard/mixed strategies.

7. What to include in the final report

- **Methodology:** FB15k-237, filtered evaluation, baseline KGE models, two-stage negative sampling.
- **Experimental setup:** hyperparameters, pool size, k , loss, seed, early stopping, validation criterion, hardware.
- **Results:** baseline table, custom strategy ablation, learning curves, slice evaluation, qualitative ranks.
- **Discussion/limitations:** RotatE-only, fixed pool size, PyKEEN/custom comparability, head-tail asymmetry.
- **Member contributions:** keep the current specific section.

8. What to include in the poster

Use a pipeline diagram, one main MRR/Hits table, one slice-evaluation chart, and one qualitative rank example. The key takeaway should be visible in one sentence: hard or mixed negatives improve ranking, especially on difficult relation/entity slices.